

Empirical ERDS Characterization — Motor Imagery

Generated `mi_erds_20260806T230536Z`. Dataset: **PhysioNet EEGMMIDB** (motor imagery, runs [4, 8, 12]), subjects [1, 2, 3, 4, 5, 6, 7, 8, 9, 10] (n=10).

Method (canonical). MNE ERDS-maps recipe (Pfurtscheller & Lopes da Silva 1999): per-epoch **multitaper** TFR (freqs 2–35 Hz, `n_cycles = freqs`), **percent-change** baseline (−1→0 s), `decim=3`. Read-outs: ERDS time-frequency maps at C3/Cz/C4 and mu-band (8–13 Hz) topographies over the 0.5–3.5 s task window. Preprocessing matches EEGPT (average ref, 0–38 Hz, 256 Hz). Convention: **percent < 0 = ERD** (desync).

Result — mu lateralization (group, C3 – C4). - LEFT imagery (T1): **+0.7%** (expect > 0 → contralateral C4 desync) - RIGHT imagery (T2): **-8.9%** (expect < 0 → contralateral C3 desync)

Central sensorimotor mu ERD is present during imagery; the ERDS maps show the mu/beta suppression time course at C3/Cz/C4. Clean bilateral-mirror lateralization sharpens with n and would sharpen further with spatial filtering / significance testing (see `docs/findings-and-options.md`). This is the empirical reference for later model comparison.

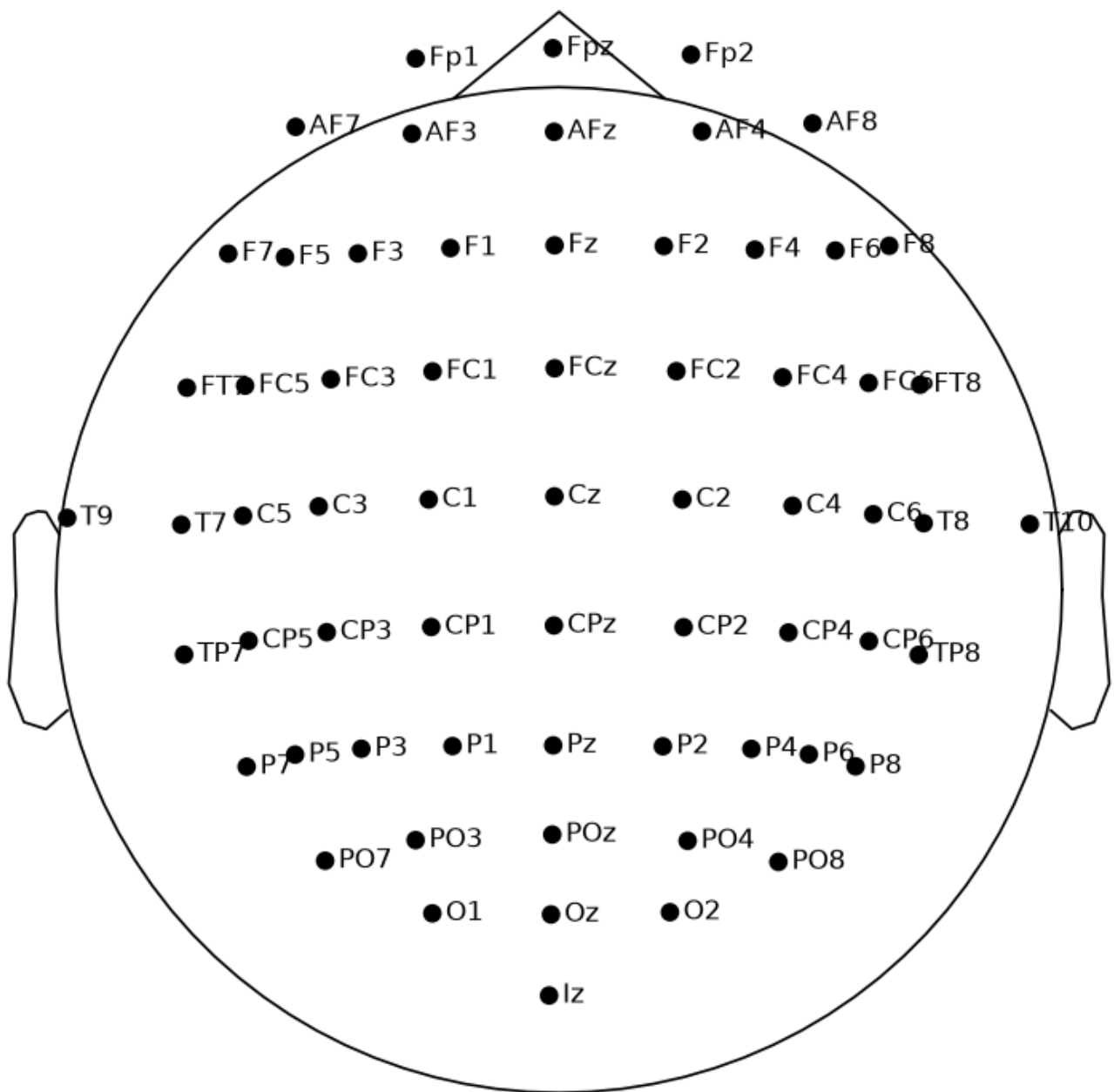
Artifacts: `qc.csv`, `erd_mu.csv`, `erds.npz`, `figures`; `config.json` records exact parameters and package versions.

Per-subject QC

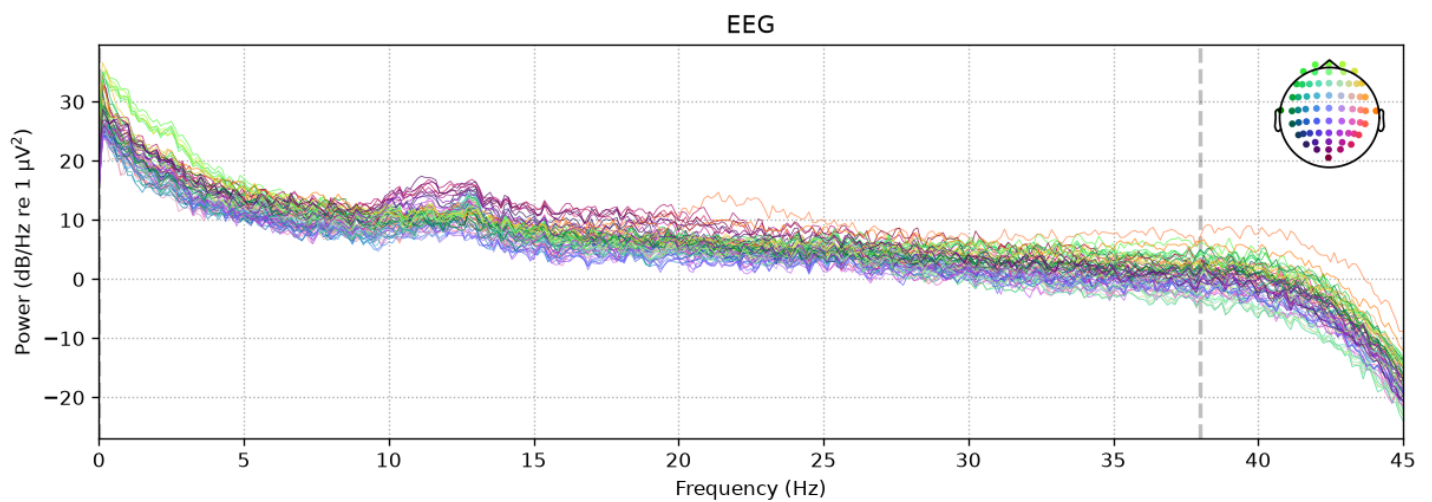
subject	sfreq	n_channels	T1	T2	high_amp_pct	ok
1	256.0	64	23	22	0.33	True
2	256.0	64	23	22	0.07	True
3	256.0	64	23	22	2.53	True
4	256.0	64	23	22	0.99	True
5	256.0	64	21	24	0.05	True
6	256.0	64	24	21	0.04	True
7	256.0	64	23	22	0.28	True
8	256.0	64	22	23	0.29	True
9	256.0	64	24	21	5.52	True
10	256.0	64	24	21	11.7	True

Figures

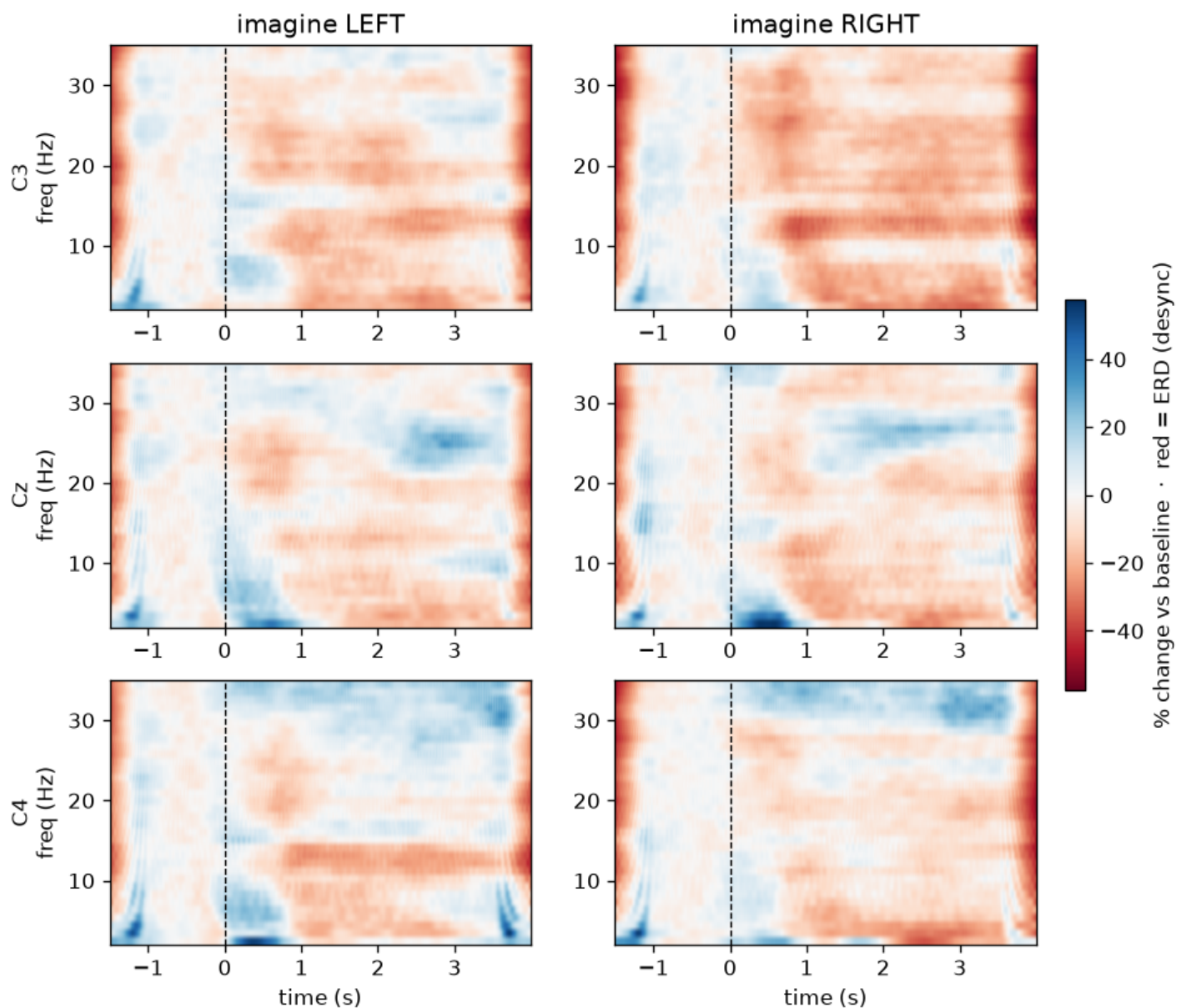
Sensor montage (64-ch, 10-10)



Power spectral density (0–45 Hz)



ERDS maps at C3/Cz/C4 (group; red = ERD)



Mu-band ERD topography — group mean (red = ERD)

